

MTH102

Linear Algebra

2025–2026–I SEMESTER

PROBLEM SETS

11 September 2025

Chapter 1. Problem set 1

Exercise 1.1. Determine the elements of the following sets.

- $\{x \in \mathbb{N} : x^2 - 1 = 0\}$.
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}$.

Exercise 1.2. What does it mean to say that A is not a subset of B ?

Exercise 1.3. Show that if A is a subset of B and B is a subset of C , then A is a subset of C .

Exercise 1.4. Show that no element n of $\mathbb{N}$ satisfies $n^4 - 5n^2 + 6 = 0$.

Exercise 1.5. What does it mean to say that two sets A, B are not equal?

Exercise 1.6. Consider the sets

$$\{1, 2, 3, 4\}, \{3, 4, 5, 6\}.$$

Determine the elements common to these sets.

Exercise 1.7. Determine the union of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.8. If A, B are sets, show that

$$A \subseteq A \cup B, B \subseteq A \cup B.$$

Exercise 1.9. If A, B are subsets of a set C , then $A \cup B$ is a subset of C .

Exercise 1.10. If A is a set, show that

$$A \cup A = A.$$

Exercise 1.11 (Commutative property). If A, B are sets, show that

$$A \cup B = B \cup A.$$

Exercise 1.12 (Associative property). If A, B, C are sets, show that

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

Exercise 1.13. Determine the intersection of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.14. If A, B are sets, show that

$$A \cap B \subseteq A, A \cap B \subseteq B.$$

Exercise 1.15. If A, B, C are sets satisfying

$$C \subseteq A, C \subseteq B,$$

then show that

$$C \subseteq A \cap B$$

holds.

Exercise 1.16. Identify the integers among

$$1, 2, 3, 4, \dots, 20$$

which are divisible by at least one of 2 and 5.

Exercise 1.17 (Inclusion-exclusion principle). Show that for finite subsets A, B, C of a set X ,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Exercise 1.18. Determine the number of integers among

$$1, 2, 3, 4, \dots, 100,$$

which are divisible by at least one of 6, 10, 15.

Exercise 1.19. If A is a set, show that

$$A \cap A = A.$$

Exercise 1.20 (Commutative property). If A, B are sets, show that

$$A \cap B = B \cap A.$$

Exercise 1.21 (Associative property). If A, B, C are sets, show that

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C).$$

Exercise 1.22 (Distributive property). If A, B, C are sets, show that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Exercise 1.23. Show that

$$A^c = \{x \in X : x \notin A\},$$

where X denotes the underlying universal set.

Exercise 1.24. Determine the complement of $\{1, 2, 3, 4\}$ in $\{1, 3, 5, 7\}$, and the complement of $\{1, 3, 5, 7\}$ in $\{1, 2, 3, 4\}$.

Exercise 1.25. If A, B are subsets of a set X , show that

$$A^c \cap B = B \setminus A.$$

Exercise 1.26. If A is a subset of a set X , then show that

$$A \cup A^c = X, A \cap A^c = \emptyset, (A^c)^c = A.$$

Exercise 1.27. If A, B are subsets of a set X , then show that

$$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c.$$

Exercise 1.28. Show that for any sets A, B , the following are equivalent.

$$\begin{aligned} A &\subseteq B, \\ A \cap B &= A, \\ A \cup B &= B, \\ B^c &\subseteq A^c. \end{aligned}$$

Exercise 1.29. If A, B, C are sets, show that

1. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C),$
2. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$

Exercise 1.30. Determine the symmetric difference of $\{1, 2, 3, 4\}$ and $\{1, 3, 5, 7\}$.

Exercise 1.31. If A, B are sets, then show that

$$\begin{aligned} A \Delta B &= B \Delta A, \\ A \cup B &= (A \Delta B) \cup (A \cap B), \end{aligned}$$

and

$$(A \Delta B) \cap (A \cap B) = \emptyset.$$

Exercise 1.32. Do there exist three finite sets A, B, C satisfying

$$|A \Delta B| = 1, |B \Delta C| = 2, |C \Delta A| = 4?$$

Exercise 1.33. If A, B are finite sets, then show that

$$|A \times B| = |A| \times |B|,$$

where for a set X , its number of elements is denoted by $|X|$.

Exercise 1.34. Determine the cartesian product of $[1, 2]$ and $[3, 4] \cup [5, 6]$.

Exercise 1.35.

Chapter 2. Induction principles

Exercise 2.1. If A is a nonempty subset of $\mathbb{N}$, and a_1, a_2 are least elements of A , then show that $a_1 = a_2$.

Exercise 2.2 (Infinitude of primes, by Saidak). Let $a_1 = 2$, and $a_{n+1} = a_n(a_n + 1)$ for any positive integer n . For any $n \in \mathbb{N}$, show that a_n has at least n distinct prime divisors.

Exercise 2.3. Show that there are infinitely many prime numbers of the form $4k + 3$, where k is a nonnegative integer.

Exercise 2.4. Show that

$$2^{2^n} - 1$$

has at least n distinct prime divisors for all $n \in \mathbb{N}$.

Exercise 2.5. Show that every positive integer greater than 1 is a prime or is a product of prime numbers.

Exercise 2.6. Let S be a finite set of size n . Show that the number of subsets of S is 2^n .

Exercise 2.7. Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.8. Show that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.9. Show that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

for all $n \in \mathbb{N}$.

Exercise 2.10. Show that

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n$$

for all $n \in \mathbb{N}$.

Exercise 2.11. Show that

$$1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}$$

for all $n \in \mathbb{N}$.

Exercise 2.12. Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Exercise 2.13. Show that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

hold for any positive integer n .

Exercise 2.14. Show that

$$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n}$$

for all $n \in \mathbb{N}$.

Exercise 2.15. Show that $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

Exercise 2.16. Show that $5^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Exercise 2.17. Show that $5^n - 4n - 1$ is divisible by 16 for all $n \in \mathbb{N}$.

Exercise 2.18. Show that $6^n - 5n - 1$ is divisible by 25 for all $n \in \mathbb{N}$.

Exercise 2.19. Show that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.

Exercise 2.20. Show that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all $n \in \mathbb{N}$ satisfying $n > 1$.

Exercise 2.21. Show that $3^n \geq n^2$ for all $n \in \mathbb{N}$.

Exercise 2.22. Show that $n! > 2^n$ for all $n \in \mathbb{N}$ satisfying $n \geq 4$.

Exercise 2.23. Show that

$$\left(1 + \frac{1}{1^3}\right) \left(1 + \frac{1}{2^3}\right) \left(1 + \frac{1}{3^3}\right) \cdots \left(1 + \frac{1}{n^3}\right) < 3$$

holds for all $n \in \mathbb{N}$.

i Remark

Does taking $P(n)$ to be the statement that

$$\left(1 + \frac{1}{1^3}\right)\left(1 + \frac{1}{2^3}\right)\left(1 + \frac{1}{3^3}\right)\cdots\left(1 + \frac{1}{n^3}\right) < 3$$

help?

Exercise 2.24. Show that

$$1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{n^3} < \frac{3}{2}$$

for all $n \in \mathbb{N}$.

Exercise 2.25. Show that

$$\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{2^2}\right)\left(1 + \frac{1}{2^3}\right)\cdots\left(1 + \frac{1}{2^n}\right) < \frac{5}{2}$$

holds for all $n \in \mathbb{N}$.

Here are a few exercises where strong induction is useful.

Exercise 2.26. Let the integers $x_1, x_2, \dots$ be defined by

$$x_1 := 1,$$

$$x_2 := 2,$$

$$x_{n+2} := \frac{1}{2}(x_n + x_{n+1})$$

for all $n \in \mathbb{N}$. Show that $1 \leq x_n \leq 2$ for all $n \in \mathbb{N}$.

Exercise 2.27. The Fibonacci sequence $F_0, F_1, F_2, \dots$ is defined by

$$F_0 := 0,$$

$$F_1 := 1,$$

$$F_2 := 1,$$

$$F_{n+2} := F_n + F_{n+1}$$

for all $n \in \mathbb{N}$. Show that

$$F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right)$$

for all $n \in \mathbb{N} \cup \{0\}$.

Exercise 2.28. Show that

$$F_0^2 + F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}$$

for all $n \in \mathbb{N} \cup \{0\}$.